Open Source Gonioreflectometer: a four-axis BRDF instrument and 3D scanner built from printed parts without soldering

Aleksander Norman[a,*]

[a] *California State University, Dominguez Hills, Carson, CA, USA*

Abstract

The Open Source Gonioreflectometer measures the bidirectional reflectance distribution function (BRDF) of flat samples and scans objects in 3D. A light arm on a 305 mm ring and three sample axes (yaw, tilt, spin) set every illumination and viewing direction relative to a fixed camera, following Ansari-Asl et al. (2024). All 42 structural parts are 3D printed in ABS; the 61 bought parts cost US$576.06 from three distributors. No soldering, drilling, sawing or crimping is needed. Calibration uses a mirror for angles, an exposure series for camera response and a pressed-PTFE tile for absolute reflectance. Fitted Lambert + GGX parameters export to game engines. **[TBD: repeatability, accuracy vs. reference data]**.

Keywords: BRDF, gonioreflectometer, material appearance, open hardware, 3D printing, photometric calibration

*Corresponding author: **[TBD: email]**

Specifications table

Hardware name	Open Source Gonioreflectometer
Subject area	Engineering and materials science; computer graphics (material appearance)
Hardware type	Measuring physical properties and in-lab sensors; imaging tools
Closest commercial analog	Laboratory gonioreflectometers and goniospectrophotometers
Open source license	Hardware CERN-OHL-S-2.0; software GPL-3.0-or-later; documentation CC BY-SA 4.0
Cost of hardware	US$576.06 in parts (cap US$630); tools US$417.80
Source file repository	https://github.com/Normansrule/Gonioreflectometer-Open-Source (DOI at release)

1. Hardware in context

Gonioreflectometers measure how a surface reflects light as a function of incident and reflected direction (Nicodemus et al., 1977). Laboratory instruments reach high accuracy at high cost (Höpe et al., 2012, 2006; Saha et al., 2024); image-based (Marschner et al., 2000; Matusik et al., 2003) and robot-arm (Lyngby et al., 2019) setups lower the cost but still need machining or industrial arms. Consumer-hardware capture (Filip et al., 2014) trades angular coverage for speed. Ansari-Asl et al. (2024) describe a multispectral goniometric device with a light arm and a three-axis sample stage. The Open Source Gonioreflectometer reproduces that geometry with printed parts, stepper motors and a USB camera, for builders without a workshop, consistent with open-hardware practice (Bonvoisin et al., 2017; Pearce, 2012).

2. Hardware description

- **Geometry.** M4 (light arm, 20–324°, 24:1 belt on the ring), M3 (yaw, ±180°, 5:1), M2 (tilt, ±90°, 5:1), M1 (spin, direct drive). Camera fixed at 350 mm; light at 300 mm (Fig. 2). Anisotropic materials need M1 (Ward, 1992).

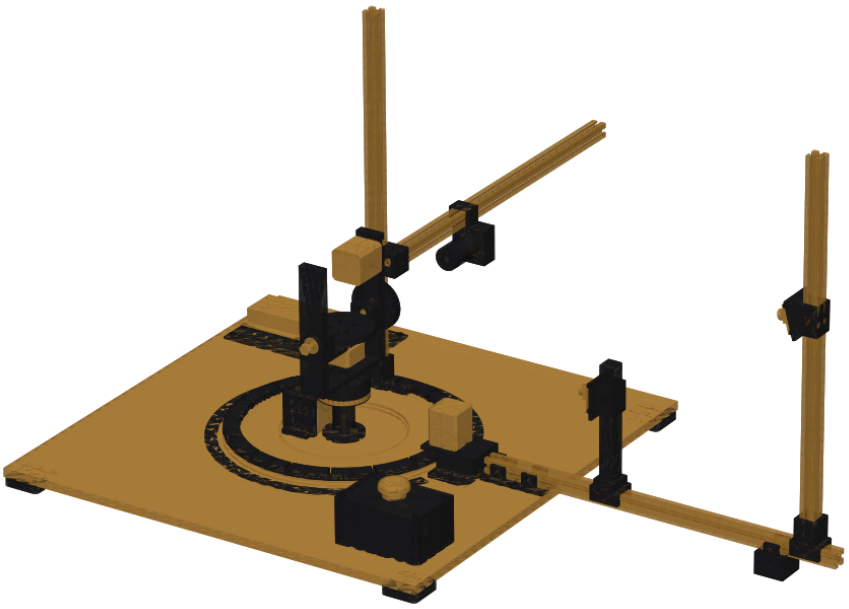


Figure 1: Whole-machine assembly from the repository CAD (`cad/assembly`). Sample axes at centre, light arm on the ring, measuring camera on the right.

- **Light.** Chip-on-board LED, colour rendering index 97, 350 mA constant current, collimating printed tube with a 10 mm aperture; solderless holder clamped by the lamp body.
- **Camera.** 12 MP IMX477 USB module, 16 mm lens, printed f/8 stop; exposure, gain and white balance locked.
- **Control.** FluidNC on an ESP32 board drives four steppers; a microcontroller reads keypads and sensors; optical homing on every axis; over-travel switches in the emergency-stop loop.
- **No-tool joins.** Lever connectors, screw terminals, a breadboard, self-tapping screws into printed holes, and wood screws without pilot holes.

3. Design files summary

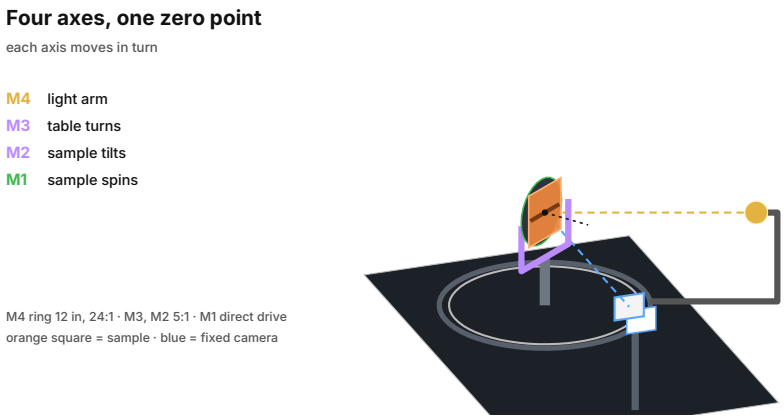


Figure 2: Four axes, one zero point. M4: light arm; M3: table yaw; M2: tilt; M1: spin.

Table 1: Design files summary. Printed parts: `phases/<stage>/3d/`, STEP and STL.

Design file	Type	License	Location
PLAN-01 tolerance coupon	STEP, STL	CERN-OHL-S-2.0	`00-plan`
PWR-18a power box	STEP, STL	CERN-OHL-S-2.0	`01-power`
PWR-18b power lid	STEP, STL	CERN-OHL-S-2.0	`01-power`
CTL-10a button panel	STEP, STL	CERN-OHL-S-2.0	`02-controller`
CTL-10b controller tray	STEP, STL	CERN-OHL-S-2.0	`02-controller`
FRM-09a og40 adapter	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-09b foot	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-09c protractor segment	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-09d extrusion saddle	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-09e post socket	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-09f end foot	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-09g washers	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-09h camera pillar	STEP, STL	CERN-OHL-S-2.0	`03-frame`
LAR-08a rim segment	STEP, STL	CERN-OHL-S-2.0	`04-light-arm`
LAR-08b ring motor mount	STEP, STL	CERN-OHL-S-2.0	`04-light-arm`
LAR-08c arm segment	STEP, STL	CERN-OHL-S-2.0	`04-light-arm`
LAR-08e cross clamp	STEP, STL	CERN-OHL-S-2.0	`04-light-arm`
TBL-07a table drum	STEP, STL	CERN-OHL-S-2.0	`05-table`
TBL-07b home flag	STEP, STL	CERN-OHL-S-2.0	`05-table`

Design file	Type	License	Location
TBL-07c table motor bracket	STEP, STL	CERN-OHL-S-2.0	`05-table`
TBL-07d table post	STEP, STL	CERN-OHL-S-2.0	`05-table`
TBL-07f spacers	STEP, STL	CERN-OHL-S-2.0	`05-table`
CAM-03a aperture stop f8	STEP, STL	CERN-OHL-S-2.0	`06-camera-light`
CAM-03b camera plate	STEP, STL	CERN-OHL-S-2.0	`06-camera-light`
LSR-05a lamp tube	STEP, STL	CERN-OHL-S-2.0	`06-camera-light`
LSR-05b lamp cap	STEP, STL	CERN-OHL-S-2.0	`06-camera-light`
CAL-06a ptfe holder	STEP, STL	CERN-OHL-S-2.0	`07-calibrate`
CAL-06b mirror holder	STEP, STL	CERN-OHL-S-2.0	`07-calibrate`
CAL-06c ball holder	STEP, STL	CERN-OHL-S-2.0	`07-calibrate`
ENC-04a dowel socket	STEP, STL	CERN-OHL-S-2.0	`07-calibrate`
TLT-07a tilt yoke	STEP, STL	CERN-OHL-S-2.0	`08-tilt`
TLT-07b pulley 100T	STEP, STL	CERN-OHL-S-2.0	`08-tilt`
TLT-07c tilt cradle	STEP, STL	CERN-OHL-S-2.0	`08-tilt`
SPN-07a spin platter	STEP, STL	CERN-OHL-S-2.0	`09-spin`
MSK-06a led ring	STEP, STL	CERN-OHL-S-2.0	`10-color`
MSK-06b led mast	STEP, STL	CERN-OHL-S-2.0	`10-color`
SPC-03a spectral tube	STEP, STL	CERN-OHL-S-2.0	`10-color`
SCN-03a riser 20	STEP, STL	CERN-OHL-S-2.0	`11-scan-capture`
SCN-03b riser 40	STEP, STL	CERN-OHL-S-2.0	`11-scan-capture`
SCN-03c riser 60	STEP, STL	CERN-OHL-S-2.0	`11-scan-capture`
SCN-03d scan camera plate 15	STEP, STL	CERN-OHL-S-2.0	`11-scan-capture`
STA-07a pi zero mount	STEP, STL	CERN-OHL-S-2.0	`15-standalone`
Bought-part stand-ins	STEP, STL	CERN-OHL-S-2.0	`cad/bought`
Whole-machine assembly	STEP	CERN-OHL-S-2.0	`cad/assembly`
CAD source (CadQuery)	Python	GPL-3.0-or-later	`tools/cad`

4. Bill of materials summary

Parts from Digi-Key, Amazon and Adafruit. Tools (US$417.80, including the printer) are listed separately in the repository. Printed parts: ≈1033 g ABS.

Table 2: Bill of materials summary (core machine; shared packs counted once; prices in USD, checked on the store pages).

Designator	Component	No.	Unit	Total	Source
FIL-01	ABS filament 1.75 mm, black, 1 kg (2 spools)	2	15.99	31.98	Amazon
PWR-01	AC adapter 24 V 2.5 A 60 W	1	35.26	35.26	Digi-Key
PWR-02	IEC C13 mains cord	1	3.14	3.14	Amazon
PWR-03	DC jack to screw-terminal adapter 2.1 mm	1	2.00	2.00	Adafruit
PWR-04	Inline 5 x 20 mm fuse holder	1	4.00	4.00	Digi-Key
PWR-05	Fuse 3.15 A time-lag 5 x 20 mm	2	1.56	3.12	Digi-Key
PWR-10	Optocoupler LTV-817 / PC817 (DIP-4)	1	0.39	0.39	Digi-Key
PWR-11	Resistors 10k (2)	2	0.10	0.20	Digi-Key
PWR-13	Emergency stop 22 mm latching, 1 NC	1	30.87	30.87	Digi-Key
PWR-15	18 AWG silicone wire red/black, 1 m pair	1	7.98	7.98	Amazon
PWR-19	Half-size breadboard (400 points)	1	4.95	4.95	Adafruit
PWR-20	Lever wire connectors, WAGO 221 assortment (12)	1	12.45	12.45	Amazon
CTL-01	FYSETC E4 (ESP32, 4x TMC2209, WiFi + Bluetooth)	1	40.99	40.99	Amazon
CTL-02	Raspberry Pi Pico H (headers soldered)	1	5.00	5.00	Digi-Key
CTL-03	USB A-to-B cable for the E4	1	3.95	3.95	Adafruit
CTL-04	USB-A to micro-B data cable 1 m	1	2.95	2.95	Adafruit
CTL-05	Membrane keypad 1 x 4	2	2.95	5.90	Adafruit
CTL-06	JST-XH pre-crimped lead kit	1	10.99	10.99	Amazon
CTL-07	Jumper wires male-female 150 mm (20)	2	1.95	3.90	Adafruit
CTL-11	Stepper cables 1 m, PH2.0 6-pin to XH2.54 4-pin (4-pack)	1	6.00	6.00	Amazon
FRM-01	Baltic birch plywood 1/2 in, 24 x 24 in	1	20.00	20.00	Amazon
FRM-02	2020 V-slot 500 mm (4-pack)	1	22.39	22.39	Amazon
FRM-03	M5 T-nuts for 2020, 50 drop-in + 50 sliding	1	5.99	5.99	Amazon

Designator	Component	No.	Unit	Total	Source
FRM-06	Socket screw kit M2–M5, 8/10/16/20 mm, with nuts	1	9.99	9.99	Amazon
FRM-08	608-2RS bearings (10)	1	4.99	4.99	Amazon
FRM-12	Wood screws #8 x 5/8 in, Type 17 point (100)	1	8.60	8.60	Amazon
LAR-01	12 in lazy Susan bearing	1	13.99	13.99	Amazon
LAR-02	NEMA 17 stepper 59 Ncm 2 A (3-pack: M4, M3, M2)	1	32.99	32.99	Amazon
LAR-03	GT2 6 mm open belt (5 m)	1	7.59	7.59	Amazon
LAR-04	GT2 20T pulley 5 mm bore (5-pack)	1	3.99	3.99	Amazon
LAR-05	Optical endstop module, slot type (6-pack)	1	8.99	8.99	Amazon
LAR-06	Roller-lever limit switch, screw terminals (2)	1	6.99	6.99	Amazon
TBL-02	GT2 6 mm closed belt 232 mm	1	8.19	8.19	Amazon
TBL-05	M8 x 75 hex bolt, partial thread + M8 nut (fixed table axle)	1	15.37	15.37	Amazon
CAM-01	USB camera 12 MP IMX477, M12 mount (UVC)	1	59.99	59.99	Amazon
CAM-02	M12 lens 16 mm, $\geq$ 1/2 in format, low distortion	1	30.11	30.11	Amazon
LSR-06	Solderless COB holder 13.5 mm (TE LUMAWISE Z35)	1	3.50	3.50	Digi-Key
CAM-07	M2.5 x 8 mm screws (100-pack)	1	6.29	6.29	Amazon
LSR-01	COB LED CRI 97, 12 V, 13.5 mm (Citizen CLU7A3)	1	5.91	5.91	Digi-Key
LSR-02	Constant-current driver 350 mA	1	5.41	5.41	Digi-Key
LSR-03	Heatsink 40 x 40 mm (from a pack)	1	6.99	6.99	Amazon
ENC-02	Black felt sheet	1	6.99	6.99	Amazon
CAL-01	PTFE sheets 100 x 100 x 3 mm (2): white tile = 3 stacked 50 mm squares	1	16.99	16.99	Amazon
CAL-02	Glass mirror tile ~50 mm	1	11.49	11.49	Amazon
CAL-04	ChArUco print + picture-frame glass	1	4.31	4.31	Amazon
TLT-02	GT2 6 mm closed belt 280 mm	1	7.49	7.49	Amazon
TLT-05	M8 x 35 hex bolts, half thread (10-pack; 2 used)	1	9.19	9.19	Amazon

Designator	Component	No.	Unit	Total	Source
SPN-01	NEMA 17 pancake stepper 16 Ncm 1 A (direct drive)	1	11.99	11.99	Amazon
SCN-01	Mounting putty 2 oz	1	2.33	2.33	Amazon
SCN-02	ChArUco scanning mat print	1	1.00	1.00	print shop / library (servic
			Total	**576.06**	

5. Build instructions

The repository splits the build into stages (power, controller, frame, each axis, camera and light, calibration). Each step lists time, parts, actions, a measured pass criterion and what to check on failure (`phases/*/README.md`).

1. Print the tolerance coupon; set clearance and self-tap hole size.
2. Wire the power box; check polarity and fuse before the first power-on.
3. Mount the controller; move one motor from the laptop.
4. Place the base mounts with the 1:1 template; assemble frame and axes; home each axis.
5. Mount camera and lamp; run the mirror test; calibrate (Section 7).

6. Operation instructions

1. `opengonio check-pose M4=60 M3=30`: angle and collision check.
2. `opengonio capture inplane -step 10 -out data/s`: in-plane sweep.
3. `opengonio curve data/s`: calibrated BRDF curve and Lambert + GGX fit (Trowbridge & Reitz, 1975; Walter et al., 2007).

Safety: the lamp is viewed only through the sample; the emergency stop cuts motor power.

7. Validation and characterization

Calibration.. Angle offset from the mirror peak ($M_3 = M_4/2$ in-plane); dark level from a lamp-off frame; camera response from an exposure series (Debevec & Malik, 1997); absolute scale from pressed PTFE (Weidner & Hsia, 1981):

$$f_r = \frac{C_{\text{lin}}}{C_{w,\text{lin}}} \frac{\rho_w}{\pi} \frac{\cos 45^\circ}{\cos \theta_i}, \qquad C_{\text{lin}} = (C - C_d)^\gamma. \tag{1}$$

Results.. **[TBD: angular repeatability per axis (mirror, 10 repeats); lamp drift over a run; PTFE reflectance vs. reference; matte and glossy samples vs. MERL (Matusik et al., 2003) or RGL ("RGL-EPFL material database," n.d.) data; scan accuracy]**.

Ethics statements

No human or animal subjects.

CRediT author statement

Aleksander Norman: Conceptualization, Methodology, Software, Hardware, Writing – original draft.

Declaration of competing interest

None.

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